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CUSTOMIZED PING COMMAND TO TEST SERVER CONNECTIVITY

Aim:

To design a customized ping program that checks the connectivity between a client and a server by sending packets and measuring the response time.

Algorithm:

- 1.Start the program and input the server IP or hostname.
- 2.Send a test packet (ICMP or TCP) to the server.
- 3.Wait for a reply within a specified timeout.
- 4.If a reply is received, calculate and display the round-trip time (RTT).
- 5.If no reply is received, display "Request timed out."
- 6.Repeat the process for a given number of times.
- 7.Display summary results (packets sent, received, lost, and average RTT).

PROGRAM:

```
import socket
```

```
import time
```

```
host = "google.com" # you can change
```

```
this port = 80      # HTTP port
```

```
count = 4           # number of pings
```

```
for i in range(count):
    try:
        s = socket.socket()
        start = time.time()
        s.connect((host, port))
        end = time.time()
        s.close()
        print(f"Reply from {host}: time={{(end-start)*1000:.2f}
ms")
    except Exception:
        print("Request timed out")
if times:
    print("\nMin RTT =", min(times), "ms")
    print("Max RTT =", max(times), "ms")
    print("Avg RTT =", sum(times)/len(times), "ms")
```

OUTPUT:

```
Microsoft Windows [Version 10.0.26100.1457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>cd C:\Users\user\OneDrive\Documents

C:\Users\user\OneDrive\Documents>python ping.py
Reply from google.com: time=168.85 ms
Reply from google.com: time=74.19 ms
Reply from google.com: time=77.56 ms
Reply from google.com: time=98.64 ms

Min RTT = 74.18704032897949 ms
Max RTT = 168.84708404541016 ms
Avg RTT = 104.81005907058716 ms

C:\Users\user\OneDrive\Documents>|
```

RESULT:

Therefore, customized ping commands was implemented and server connectivity was tested.